

Laura Inno

PERSONAL INFORMATION

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- **ORCID:** 0000-0002-0271-2664
- **Current Position:**
Associate Professor of Astrophysics, SD FIS/05, Department of Science and Technology, University of Naples “Parthenope”
Italian National Scientific Qualification for Associate Professor,
SD FIS/05, obtained in May 2022

RESEARCH EXPERIENCE

- My research focuses on time-domain astronomy, large astronomical surveys, Solar System science, exoplanets, and stellar populations. I currently work on the automated detection and characterisation of cometary activity in Rubin LSST data and on the identification of suitable targets for the ESA *Comet Interceptor* mission. My research combines astronomical image analysis, time-series methods, machine learning, and physically motivated modelling of cometary dust environments.
I have written, or contributed to, more than 50 refereed publications, conference proceedings, white papers, community reports, and research notes. My ADS **h-index** is currently about **23**, with more than **1500** citations; these values should be refreshed from ADS before formal submission.

Orcid ID: 0000-0002-0271-2664.

01/07/2025–TO
DATE

Associate Professor of Astrophysics

Science and Technology Department, University of Naples “Parthenope”, Centro Direzionale Isola C4 - 80143 Naples

- Associate Professor in the scientific-disciplinary sector FIS/05.
- European Space Agency appointed member of the *Comet Interceptor* Target Identification Working Group.
- Co-Investigator of DISC, the Dust Impact Sensor and Counter onboard the ESA *Comet Interceptor* mission.
- Responsible scientist / PI of WP 5300, “Activity in support of Comet Interceptor target selection”, within ASI-INAF agreements supporting the Italian participation in *Comet Interceptor*.
- Contribution leader of the approved Italian Rubin LSST in-kind contribution ITA-INA-S11, focused on software for automated detection and characterisation of cometary activity in LSST data.
- Coordinator of TailKit / ADACTED / ComeToolKit activities for cometary dust-environment characterisation and Rubin-era Solar System science.
- Member of the Rubin Observatory LSST Science Collaborations, including the Solar System Science Collaboration.
- Member of the Research Unit of the funded PRIN project “ESPLORA: Exoplanet Spectroscopy at high resolution to Probe their Lost Origins by Revealing their Atmospheric Composition”.
- Member of observing programme / time allocation panels and referee for international astrophysical journals.
- Main teaching responsibilities include Physics, General Physics, Electromagnetism, and Physics and Quantum Computing courses at bachelor and master level.

Assistant Professor (Ricercatore a Tempo Determinato di Tipo B)

01/07/2022–
30/06/2025

Science and Technology Department, University of Naples “Parthenope”, Centro Direzionale Isola
C4 - 80143 Naples

- European Space Agency appointed member of the space mission *Comet Interceptor* Target Identification Working Group.
- Co-Investigator of the DISC instrument, part of the payload of *Comet Interceptor*.
- PI of WP 5300, “Activity in support of Comet Interceptor target selection”, as part of ASI-INAF financial agreements supporting the Italian participation in *Comet Interceptor*.
- Member of Vera C. Rubin Observatory LSST Science Collaborations, with primary involvement in Solar System science.
- Contribution leader of an approved Rubin LSST in-kind contribution, financed through the Italian Rubin/INAF framework, to gain data-rights access to LSST products.
- Founding member of the ExoplaNats collaboration, focused on identifying and characterising exoplanets from space photometric and spectroscopic data.
- Member of PETS, the PEPSI/LBT Exoplanet Transit Survey.
- Main responsible for courses in Physics and General Physics for Computer Science, Cybersecurity Engineering, and related degree programmes at the University of Naples “Parthenope”.
- Member or external collaborator of major astronomical surveys and collaborations, including SDSS, Gaia-ESO, WEAVE, and Rubin LSST.

23/08/2021–
24/01/2022

Maternity Leave

Postdoctoral researcher (Ricercatore a Tempo Determinato di Tipo A)

Science and Technology Department, University of Naples “Parthenope”, Centro Direzionale Isola
C4 - 80143 Naples

- I joined the University of Naples “Parthenope” after winning a RTDA position financed within the “Programma Operativo Nazionale Ricerca e Innovazione 2014–2020 - Fondo Sociale Europeo, Azione I.2 Attrazione e Mobilità Internazionale dei Ricercatori 2019”.
- PI of WP 5300, “Activity in support of Comet Interceptor target selection”, and co-responsible of WP 5200 within the ASI-INAF agreement n. 2020-4-HH.0 on the Italian participation in the ESA *Comet Interceptor* mission.
- Teaching activities included Planetary Environments and Astrobiology for the International PhD Programme “Environment, Resources and Sustainable Development”, General Physics, and Physics for Computer Science students.
- I supervised bachelor and master theses on machine-learning applications to astronomical data.

Supervisor: Prof.ssa Alessandra Rotundi

Department: Science and Technology

05/11/2018–
26/07/2019

Postdoctoral researcher (Assegno di Ricerca)

INAF-Osservatorio Astrofisico di Arcetri (OAA) - Florence, Italy

- In the last three months of my fellowship, I became a member of the Gaia-ESO Survey, where I was actively involved in the final stages of the Quality Control and Phase 3 delivery for the final public release, and of the WEAVE Science Team.
- During the first six months of my fellowship, I was a member of the ESA Gaia DPAC (CU-5) and part of Development Unit 13, for which I produced telluric-corrected spectra for the Standard Spectroscopic Stars Survey.

Supervisor: Dr. Sofia Randich

Department: Stars and Star Formation

01/07/2015–
31/10/2018

Postdoctoral researcher (Ricercatrice TD)

Max Planck Institut fuer Astronomie (MPIA) - Heidelberg, Germany

- Leader of a project for data mining the Pan-STARRS1 3π time-domain multi-band catalogue.
- Data mining of large ground-based survey datasets.
- Full member of SDSS and 4MOST.
- Member of the SFB-881 science collaboration.

Supervisor: Prof. Hans-Walter Rix

Department: Galaxies and Cosmology, Milky Way Group

09/05/2015–
13/06/2015

Visitor researcher

The University of Tokyo, School of Science, 7-3-1 Hongo, Bunkyo-ku, Tokyo, Japan

- I was involved in the exploitation of time-series obtained from the Japanese IRSF/SIRIUS photometric surveys.
- I joined the engineering team of the new near-infrared spectrograph WINERED, located at the Kyoto Observatory at the time and later moved to NTT.
- As a member of the Local Organising Committee, I contributed to the PhD summer school “New era of the cosmic distance scale”, Tokyo, June–July 2015.

Host: Dr. Noriyuki Matsunaga

Visitor researcher

European Southern Observatory, Garching bei Muenchen, Germany

- I carried out projects started during my PhD and concluded my work on the use of near-infrared templates for determining Cepheid distances.

Host: Dr. Martino Romaniello

Ph.D fellowship student

European Southern Observatory, Garching bei Muenchen, Germany

- During my PhD, I focused on the use of Cepheids to address open questions including the calibration of the extragalactic distance scale and the dynamical evolution of the Magellanic Clouds.
- The ESO studentship allowed me to develop research independence through observing preparation, data analysis, astrophysical interpretation, and publication.

Supervisor: Dr. Martino Romaniello

Ph.D fellowship student

University of Rome Tor Vergata, Via della Ricerca Scientifica 1, Rome, Italy

I ranked first at the admission exam for the “Scuola di Dottorato in Astronomia: XXVII ciclo”, thus obtaining the national fellowship.

- I widened my expertise by taking in-house courses and attending international PhD schools and conferences.
- I worked as a tutor for master students.
- I successfully applied to the ESO studentship programme.

Supervisor: Prof. Giuseppe Bono

EDUCATION AND TRAINING

PhD in Astronomy

Universita di Roma Tor Vergata, Rome, Italy

Thesis title: “Cepheids as stellar tracers to determine galactic structures: an application to the Magellanic Clouds”

Grade: cum Laude

Supervisors: Prof. Giuseppe Bono, Dr. Martino Romaniello

Master Degree (Laurea Specialistica in Scienze dell’Universo)

Universita di Roma Tor Vergata, Rome, Italy

Thesis title: “Distance and geometry of the Magellanic Clouds using the Cepheid Near-Infrared Period-Luminosity Relations.”

Grade: cum Laude

Bachelor Degree in Physics (Laurea triennale in Fisica - curriculum astrofisica)

Universita degli Studi di Salerno, Fisciano, Salerno, Italy

Thesis title: “The Structure of the Milky Way Bulge”

Grade: 108/110

Supervisor: Prof. Gaetano Scarpetta

RESEARCH LEADERSHIP, PROJECTS AND COLLABORATIONS

2023–TO DATE

Rubin LSST in-kind contribution ITA-INA-S11 Contribution leader for the Italian in-kind software contribution focused on automated detection and characterisation of cometary activity in Rubin LSST data. The activity includes modular pre-processing, moving-object centering and cropping, activity diagnostics, and forward modelling of cometary dust environments.

2023–TO DATE

TailKit / ADACTED / ComeToolKit Coordination of the development of software tools for cometary dust-environment characterisation in support of Rubin-era Solar System science and ESA *Comet Interceptor* target identification.

2020–TO DATE

ESA Comet Interceptor Member of the Target Identification Working Group, Co-Investigator of the DISC instrument, and responsible scientist for activities supporting target selection and dust-environment analysis.

2019–TO DATE

ExoplaNats Founding member of a collaboration involving the University of Naples Federico II, the University of Naples “Parthenope”, and NASA Goddard Space Flight Center, focused on exoplanet identification and characterisation from space photometric and spectroscopic data.

TEACHING

Main teaching responsibilities at the University of Naples “Parthenope” include:

- Physics for first-year students of the Computer Science bachelor degree.
- General Physics for first-year bachelor students of Cybersecurity Engineering and related degree programmes.
- Physics II / Electromagnetism modules for scientific, nautical, aeronautical, and meteo-oceanographic degree programmes.
- Physics and Quantum Computing for the MSc in Applied Computer Science - Big Data and Machine Learning.
- Planetary environments and astrobiology topics for the International PhD Programme “Environment, Resources and Sustainable Development”.

INTERNATIONAL CONFERENCES

During my research career, I have attended several international meetings. I have been invited to talk at several of them and have presented contributed talks and posters at many others.

2025

Invited talk at the LSST Solar System follow-up workshop, Royal Observatory Edinburgh.

2024

Contributed talk at the Europlanetary Science Congress, Berlin, Germany.

2024

Contributed talk at the European Astronomical Society Conference, Padova, Italy.

2024

Contributed talk at the VST Science Workshop, Naples, Italy.

2023

Contributed talk at LSST@Europe5, Porec, Croatia.

2023

Invited talk at the 109th National Congress of the Societa Italiana di Fisica.

2022

LSST@Europe4 - Contributed Talk.

2021

EAS 2021 - accepted contributed talk.

2021

EGU General Assembly, session on equality, diversity and inclusion in the geosciences.

2021

Invited speaker for “Giornata internazionale delle Donne e delle Ragazze nella Scienza”, Edu-INAF.

2020

EAS 2020 - invited talk for the session “The Magellanic Clouds in the 2020s: a spectroscopic perspective”.

2019

European Week of Astronomy and Space Science, Lyon - three accepted contributed talks.

2019

Lorentz Center, “Metals in Galaxies, Near and Far: Looking Ahead” - invited speaker.

2019

Sexten Center for Astrophysics, “Cosmic Beacons” - invited speaker.

2018

IAU Focus Meeting FM7 and IAU Symposium 347, Vienna - invited/contributed talks.

2017

“Piercing the Galactic Darkness”, Heidelberg - invited/contributed talk.

2016

MIAPP workshop “The Chemical Evolution of Galaxies” - invited speaker.

2016

ISSI-Beijing workshop “Astronomical Distance Determination in the Space Age” - invited speaker.

2015

PhD School “F. Lucchin: Science and Technology with E-ELT”, Erice - invited lecturer.

2013–2014

ESO, MIAPP, La Serena and Vatican EWASS meetings - talks and posters on Cepheids, Magellanic Clouds, and the distance scale.

INVITED SEMINARS

19/05/2023

Seminar Dipartimento di Fisica, Universita degli Studi di Roma 3.

16/06/2022

	Seminar	Web-Seminar INAF-Osservatorio Astronomico d'Abruzzo.
23/05/2019		
	Seminar	INAF-Osservatorio Astrofisico di Torino.
05/12/2018		
	Astromeeeting	INAF-Osservatorio Astronomico di Capodimonte.
03/05/2018		
	Joint Colloquium	INAF-OAS Bologna and Dipartimento di Fisica e Astronomia, Universita di Bologna.
20/01/2017		
	Koenigstuhl Colloquium	MPIA, Heidelberg.

REFeree FOR ASTRONOMICAL JOURNALS

2018–TO DATE	I regularly serve as a referee for major international astrophysical journals, including ApJ, MNRAS, and Nature Astronomy. I am Review Editor for Frontiers in Physics and Frontiers in Astronomy and Space Sciences.
2020–TO DATE	I serve in telescope allocation / observing programme committees, including TNG and ESO-related panels.

CONFERENCE ORGANISATION AND GRANTS

2020–2022	Main organiser of monthly seminars hosted by the ExoplaNAts group.
2018	At INAF-OAA, organiser of the biweekly Astrobigne seminars.
2016–2017	Main organiser and SOC chair of the conference “Piercing the Galactic Darkness: Stellar populations in highly extincted regions of the Milky Way” (GALDARK2017).
2016–2017	I led the fundraising campaign and managed the conference budget; I was awarded 11,000 euro from the SFB881 Collaborative Research Center at the Zentrum fuer Astronomie der Universitaet Heidelberg.
2012–2016	Member of the organising committee for several international conferences and schools, including Tokyo “New era of the cosmic distance scale”, PhD School “F. Lucchin”, RASPUTIN, Young Researcher Meeting in Trieste, and EWASS2012.

TELESCOPE TIME AND OBSERVING EXPERIENCE

TELESCOPE TIME AWARDED	I have been successful in obtaining telescope time at highly competitive facilities. As PI, I have been awarded telescope time at ESO VLT, CAHA/CARMENES, MPG 2.2m/GROND and FEROS, LBT/LUCI, and TNG/GIANO. As Co-I, I have been awarded time at ESO VLT, ESO NTT, TNG/GIARPS, TNG/DOLORES and NICS, HST, Subaru, and IRSF/SIRIUS.
TELESCOPE OBSERVING EXPERIENCE	I have been visitor observer at ESO NTT, TNG, VLT, and OHP, with experience in optical and near-infrared spectroscopy.
ACQUIRED SKILLS FROM OBSERVING EXPERIENCE	In the context of planning and executing observations, I have developed hands-on expertise in: • selecting diagnostics to derive kinematical, chemical, and physical properties of targets; • handling photometric catalogues and databases for target selection; • using preparation tools for observing blocks; • reducing spectroscopic raw data through ESO and facility pipelines; • using software for telluric correction, spectroscopic analysis, and survey data mining.

AWARDS AND SCHOLARSHIP

2021

Nominee to apply to the USERN Prize 2021.

2019–2022

Winner of the selection for the PON Ricerca e Innovazione 2014–2020, Attraction and International Mobility, Fondo Sociale Europeo, Azione I.2.

2015–2018

Selected through the annual call for MPIA postdoctoral positions.

2012–2014

ESO PhD Fellowship.

2011–2014

Borsa di studio ministeriale Dottorato XXVII ciclo.

MEMBERSHIP OF SCIENTIFIC SOCIETY AND INTERNATIONAL COLLABORATIONS

2025–TO DATE

Member of the International Astronomical Union.

2021–TO DATE

Member of Rubin Observatory LSST Science Collaborations and Italian Rubin LSST team.

2019–TO DATE

Member of the Italian team for LSST.

2018–TO DATE

External Collaborator for SDSS.

2019–TO DATE

Member of WEAVE.

2012–TO DATE

Member of SAIIt (Società Astronomica Italiana).

2018–2020

Member of Gaia DPAC (CU5).

2016–2018

Member of the 4MOST Magellanic Clouds Survey.

2015–2018

Member of SFB 881 “The Milky Way System” and full member of SDSS.

PUBLIC OUTREACH AND SCIENCE COMMUNICATION

I am very enthusiastic about public outreach and have always been active in this field. Selected activities are listed below.

2026

Coordinator/proposer of the University outreach project “Noi, la loro eredità”, promoted by the CUG for the International Day of Women and Girls in Science.

2020–2021

National interviews and media activities:

- Interview for Agenzia Dire.
- Interview for Corriere della Sera.
- Live interview for Meeting Science.
- Live interview for She is a Scientist.
- Interview for Il Bo Live, Padova.
- Live interview for Radio Rai 3 for the International Day of Women and Girls in Science.

22/12/2020

Invited speaker to the panel discussion organised by ESERO on “STudio, Entusiasmo, AMore per le scienze”.

2019, 2021,

2022

Scientific explainer for the science festival “Futuro Remoto”, Città della Scienza, sponsored by the University of Naples “Parthenope”.

27/09/2019

Scientific explainer for SHARPER at the European Researchers’ Night in Naples.

22/02/2018

Invited public talk at the Civico Planetario Ulrico Hoepli of Milan: “Fari nel buio: le stelle pulsanti ci svelano la Galassia nascosta”.

2009–2018

Collaborator for the outreach association ScienzImpresa; ESO Italian translator and Hubble Podcast subtitle editor.

2011–2012

Tour guide for the Astronomic Museum ASTROLAB of the Astronomic Observatory of Rome.

SUPERVISING AND MENTORING

2020–TO DATE

Supervisor and co-supervisor of bachelor and master theses in Computer Science, Machine Learning, time-domain astronomy, exoplanet detection, and Solar System science at the University of Naples “Parthenope”.

2023–TO DATE

Supervisor and coordinator of students and early-career researchers involved in Rubin LSST / *Comet Interceptor* software and science activities.

OCTOBER 2015

Lecturer at the PhD School “F. Lucchin: Science and Technology with E-ELT”, where I offered tutorials on Python and IDL for time-domain analysis.

APRIL 2012–

AUGUST 2012

Tutor for the second semester of the Astromundus programme for the Stellar Astrophysics class.

PERSONAL SKILLS

MOTHER

TONGUE(S)

Italiano

FOREIGN

LANGUAGE(S)

English, German

LEVELS

A1 and A2: Basic user - B1 and B2: Independent user - C1 and C2: Proficient user

[Common European Framework of Reference for Languages](#)

Understanding		Speaking		Writing
Listening	Reading	Spoken interaction	Spoken production	
C2	C2	C2	C2	C2
A1	A1	A1	A1	A1

COMMUNICATION

SKILLS

- Good communication and interaction skills gained through experience in large collaborations, including SDSS, 4MOST, Gaia, and Rubin LSST, and through extensive outreach activities.
- Good knowledge of presentation and writing software: PowerPoint, Keynote, Prezi, TeX/LaTeX, Overleaf, and Office tools.

ORGANISATIONAL

/ MANAGERIAL

SKILLS

Excellent organisational and managerial skills, demonstrated by the organisation of GALDARK2017, ExoplaNats seminars, and several international schools, meetings, and outreach activities.

JOB-RELATED

SKILLS

During my time at ESO, MPIA, INAF, and Parthenope, I developed expertise in time-domain analysis, large-survey data mining, machine-learning methods, Bayesian/probabilistic modelling, astronomical image analysis, and spectroscopic analysis.

Coding experience:

- Python and large survey database APIs.
- IDL, IRAF, TOPCAT, R.
- SQL, MPI.
- HTML, PHP.

Autorizzo il trattamento dei dati personali contenuti nel mio curriculum vitae in base all’art. 13 del D. Lgs. 196/2003 e all’art. 13 del Regolamento UE 2016/679 relativo alla protezione delle persone fisiche con riguardo al trattamento dei dati personali.